VLSI DESIGN INTERNSHIP

TASK 5 REPORT

RTL Design of Datapath, ALU, and Memory Components (RAM/ROM)

Submitted by	T. Ruchitha
Task focus evidenced here	RAM write/read and ROM lookup simulation
Design / simulation environment	Verilog/SystemVerilog on EDA Playground using Icarus Verilog
Evidence supplied	EDA Playground console screenshot and EPWave waveform screenshot

This report incorporates the screenshots supplied from the EDA Playground run. The console output reports a successful RAM test and ROM address 5 test, while ROM address 0 reports FAIL and RAM address 0 displays unknown (xx). These results are reported as shown rather than being presented as a fully passing test suite.

1. Objective

The objective of Task 5 is to design and verify datapath and memory components using RTL. The supplied EDA Playground evidence specifically demonstrates RAM read/write activity and ROM lookup tests. The task brief also identifies ALU, register-file, datapath integration, testbench development, and waveform analysis as learning areas.

2. Tools and Simulation Setup

The screenshots show EDA Playground configured with SystemVerilog/Verilog and Icarus Verilog 12.0. The testbench and design files are named testbench.sv and design.sv in the console command. EPWave is opened after the run to inspect the generated VCD waveform.

3. Functional Overview

RAM: The console reports a write to address 3 with data A5, followed by a read from address 3 returning A5. The waveform also shows the write-enable signal, write data, address, and read data activity.
ROM: The testbench checks ROM address 0 and address 5. The displayed output shows address 0 returning xx and failing its check, while address 5 returns 65 and passes its check.

4. Simulation Console — Actual Screenshot

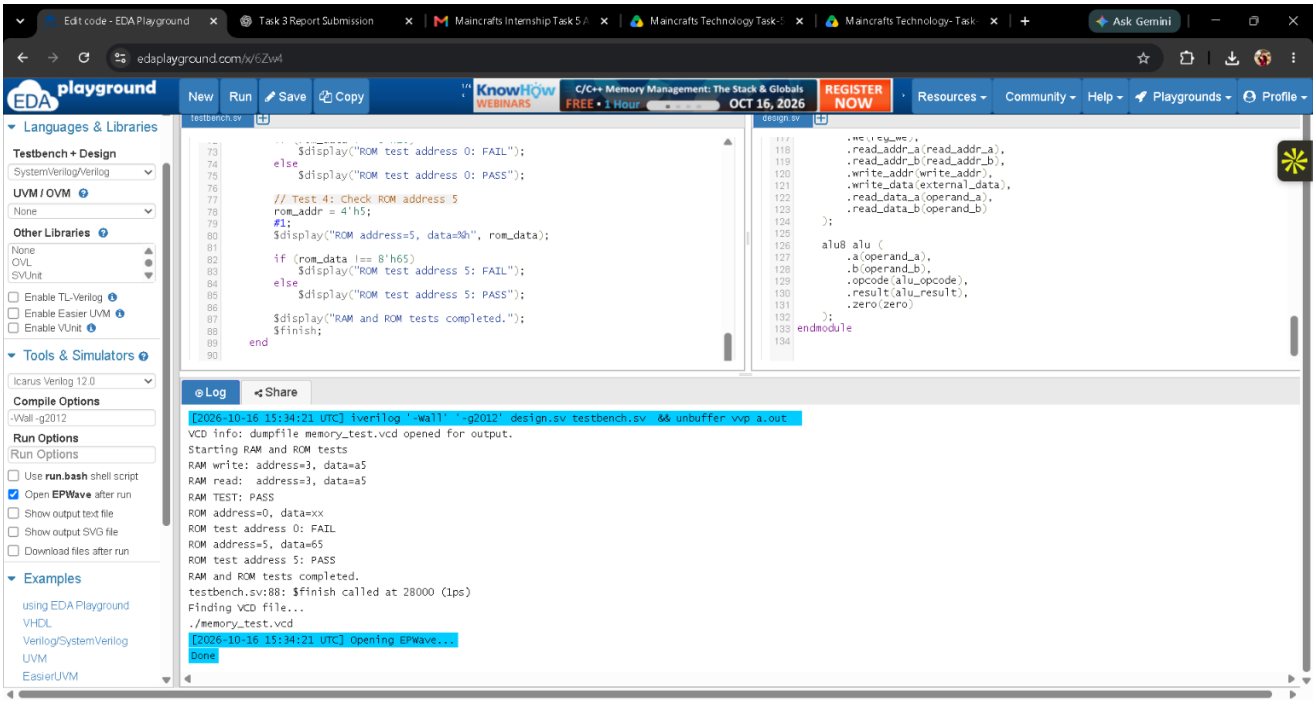


Figure 1. EDA Playground run log. The log states: RAM write address=3, data=A5; RAM read address=3, data=A5; RAM TEST: PASS; ROM address 0 data=xx and FAIL; ROM address 5 data=65 and PASS. It also shows the simulation finishing at 28,000 time units and opening EPWave.

5. EPWave Timing — Actual Screenshot

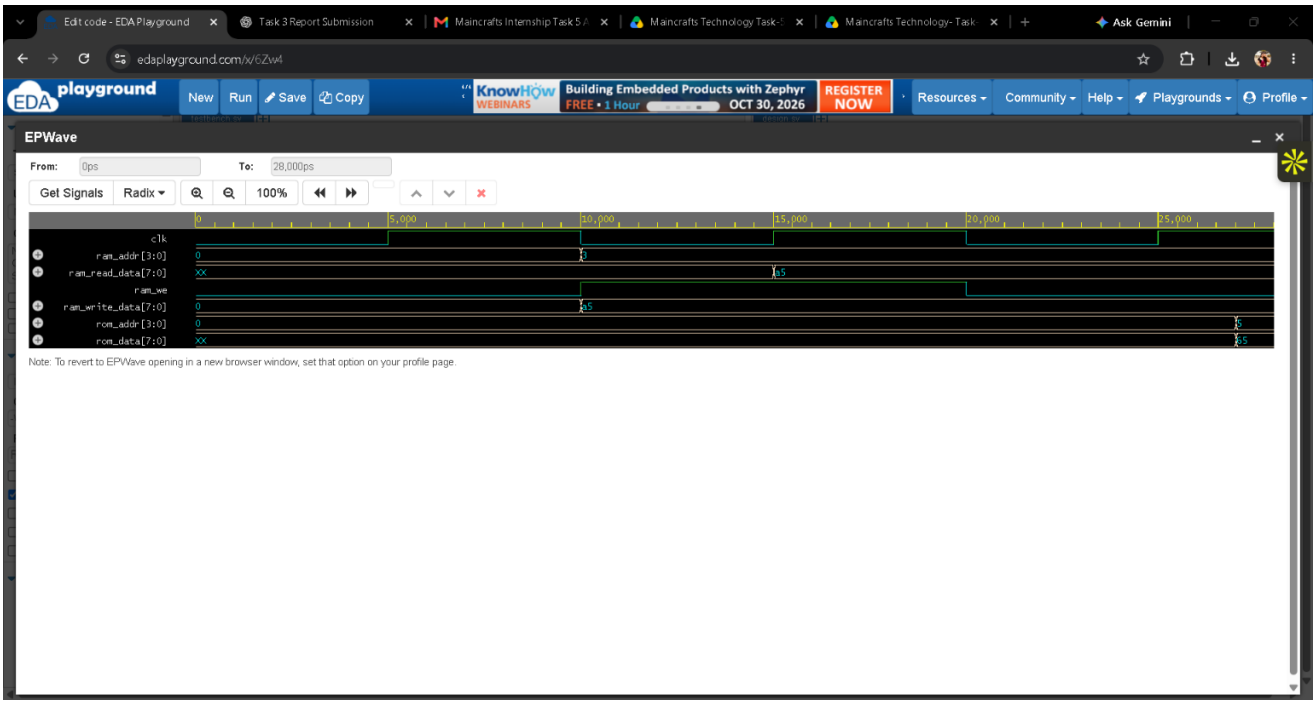


Figure 2. EPWave waveform screenshot showing clk, ram_addr, ram_read_data, ram_we, ram_write_data, rom_addr, and rom_data. The waveform includes RAM address 3 and write data A5, with the read data later showing A5; ROM address 5 and data 65 are visible near the end of the captured window.

6. Results and Interpretation

Test / observation	Evidence shown	Result
RAM write	Address 3, data A5	Reported in console
RAM read-back	Address 3 returns A5	PASS
ROM address 0	Read data is xx	FAIL
ROM address 5	Read data is 65	PASS

The RAM write/read-back behavior is consistent for address 3 in the captured run. ROM address 5 also matches the expected value checked by the testbench (8'h65). ROM address 0 remains unresolved because the observed value is unknown (xx), which causes the testbench failure. The ROM initialization or address-0 case should be inspected before describing the complete memory test as passing.

7. Recommended Debugging Steps

1. Inspect the ROM initialization or case statement and confirm that address 0 is assigned a known 8-bit value. 2. Check that the ROM address is driven before sampling rom_data. 3. Confirm the testbench comparison uses the same expected value and that any required delay or clock edge has elapsed before checking. 4. Rerun the test and verify that ROM address 0 no longer returns xx. 5. Save a new console screenshot and waveform after the correction.

8. Conclusion

The supplied EDA Playground screenshots provide evidence of a RAM write/read-back at address 3 and a successful ROM lookup at address 5. They also reveal a failing ROM address-0 test due to unknown data. This report preserves that result accurately and identifies the next debugging action instead of claiming every test passed.

Appendix — Screenshot Evidence

Both screenshots supplied by the user are embedded in this report: the EDA Playground console/log and the EPWave timing window. These are actual screenshots from the provided simulation session.